

Adelowo Adebajji Oluwatimileyin

Applied Mathematician · Machine Learning Engineer

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PROFILE

Applied Mathematician and Machine Learning Engineer with experience developing predictive models, data pipelines, signal-processing systems, and ML-backed APIs for aviation and operational applications. Strong foundation in machine learning, numerical computing, and statistical analysis, with hands-on experience in Python, PyTorch, scikit-learn, SQL, FastAPI, and Docker. Additional project experience spans LLM applications, computer vision, scientific machine learning, generative AI, and computational modelling.

PROFESSIONAL EXPERIENCE

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| Dec 2024 – Present | Data Scientist <i>Intuos Srl</i> , L'Aquila, Italy <ul style="list-style-type: none">Develop machine-learning classifiers for automated aircraft flight-phase identification from operational telemetry.Design signal-processing pipelines for aircraft engine RPM estimation, frequency analysis, and anomaly detection.Build automated telemetry-processing workflows and operational analytics for flight-performance monitoring.Develop REST APIs using FastAPI to expose machine-learning models and engineering analytics to operational systems. |
| Apr 2023 – Nov 2024 | Machine Learning Engineer <i>N-AND Italia Srl</i> , Italy <ul style="list-style-type: none">Developed predictive models for vending-machine delivery-failure forecasting using large-scale transaction and telemetry data.Performed exploratory and statistical analysis to identify patterns affecting performance and operational reliability.Built data-processing and KPI-monitoring workflows to support operational analysis and decision-making. |
| Jan 2023 – Mar 2023 | Operational Data Analyst <i>eResult Srl</i> , Cesena, Italy <ul style="list-style-type: none">Optimized SQL queries and automated recurring business-intelligence reporting workflows.Transformed operational data into performance metrics and reports for business users. |
| Apr 2022 – Oct 2022 | Digital Engineering Intern <i>Bridgestone EMIA</i> , Rome, Italy <ul style="list-style-type: none">Developed geometry-processing algorithms to support tire design, engineering simulation, and mesh generation. |
| May 2022 – Aug 2022 | Machine Learning Intern <i>Fellowship.AI</i> , Remote <ul style="list-style-type: none">Built an AI image-generation workflow and deployed model inference through FastAPI for integration with an iOS application. |

TECHNICAL SKILLS

Programming and data: Python, SQL, C++, MATLAB, JavaScript, NumPy, SciPy, Pandas, PostgreSQL, IBM DB2

Machine learning: PyTorch, TensorFlow, scikit-learn, LightGBM, supervised learning, deep learning, feature engineering, model evaluation, time-series analysis, signal processing

LLM and generative AI: LLMs, RAG, LangChain, OpenAI APIs, ChromaDB, NL-to-SQL, prompt engineering, embeddings, vector search

Software, data engineering and MLOps: FastAPI, Docker, MLflow, Apache Spark, Databricks, Git/GitHub, ETL/data pipelines, REST APIs, AWS, Kubernetes, Terraform

Mathematical computing: Statistical analysis, optimization, numerical methods, computational modelling

SELECTED PROJECTS

Flight Data Monitoring Dashboard: Built an aviation analytics platform using FastAPI, IBM DB2, React, and Docker. Integrated machine-learning classifiers and recorded flight data to support internal flight-performance analysis.

LLM and Generative AI Applications: Built nine AI applications, including a medical document-search assistant and a natural-language-to-SQL system, using technologies including LangChain, ChromaDB, model fine-tuning, and model compression.

Aircraft Engine Monitoring: Built a Raspberry Pi system that estimates engine speed and operating status from microphone audio. Connected the signal-processing pipeline to an embedded controller and tested it on a real aircraft engine.

Neural Networks for Physical Simulation: Built a physics-informed neural network in PyTorch for the 1D advection–diffusion equation, incorporating the governing PDE, initial conditions, and boundary conditions directly into training.

Medical Image Segmentation: Developed a 3D U-Net pipeline in PyTorch for liver segmentation from CT volumes, covering data preparation, model training, evaluation, and inference.

Numerical Modelling and Optimization: Developed a numerical simulation and optimization workflow for studying the mixing of passive scalars in incompressible flows, combining mathematical modelling, numerical methods, and computational analysis.

S4D vs. Transformer: Sequence-Architecture Benchmark: Implemented an S4D state-space model and a causal Transformer from scratch in PyTorch and benchmarked them under matched training conditions, reaching $4.5\times$ higher training throughput at longer context lengths.

Whole-Body Diffusion Generator: Built a face-guided image-generation pipeline using IP-Adapter FaceID and SDXL, with model inference deployed on serverless GPU infrastructure.

Photoacoustic Reconstruction: Built a synthetic photoacoustic-imaging pipeline comparing classical time-reversal reconstruction with learned U-Net refinement, improving reconstruction PSNR by 11.8 dB.

Phase-Field Modelling of Multiphase-Flow PDEs: Built a numerical PDE solver coupling a phase-field interface model with incompressible Navier–Stokes equations and heat transfer for boiling simulations, using an FFT-based pressure solve designed for 3D extensibility.

EDUCATION

2018 – 2021	M.Sc. Mathematical Engineering University of L’Aquila, Italy Final grade: 106/110 . Focus: optimization, numerical analysis, scientific computing, and applied mathematics.
2013 – 2018	B.Sc. Mathematics Obafemi Awolowo University, Nigeria First Class Honours

AWARDS AND LEADERSHIP

Awards: University of L’Aquila International Scholarship (2018–2021); Professor S.S. Okoya Prize for Excellence in Applied Mathematics (2018); MTN Foundation, PTDF, and Murli T. Chellaram Foundation scholarships (2014–2018).

Leadership: Founder and President, Inter-University Mathematics Club (2016–2018); Academic Director and peer tutor, NAMSS, Obafemi Awolowo University (2015–2018).